Albert Einstein

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Education

University of Alberta, Computer Engineering — Edmonton, Canada, 2023 – 2026

Isfahan University of Technology, Computer Engineering — Isfahan, Iran, 2018 – 2023

Experience

Very Large Scale Integration (VLSI) Research Lab, University of Alberta, Graduate Research Assistant — Edmonton, Canada, 2023 – 2026, 3 years

- Designed and implemented AI architectures, including training and inference circuits, on FPGAs.
- Integrated and synchronized IP core modules with edge detector 1-stage synchronizers.
- Wrote testbench modules for behavioral, functional, and timing simulation and verification of RTL designs.
- Defined implementation constraints including clock period to meet timing requirements.
- Optimized dynamic power by minimizing switching activity through stop conditions.
- Fine-tuned distance functions in clustering algorithms and derived gradients in backpropagation algorithms.

Structured Machine Learning Research Lab, Simon Fraser University, Graduate Research Assistant — Burnaby, Canada, 2023 – 2024, 1 year

- Counted motifs in a graph database by matrix multiplication, suitable for ML applications.

Volunteer

People's Climate March, Lead Organizer — Zurich, Switzerland, Apr 2014 – July 2015

Lead organizer for the New York City branch of the People's Climate March, the largest climate march in history.

- Awarded 'Climate Hero' award by Greenpeace for my efforts organizing the march.
- Men of the year 2014 by Time magazine

Awards

Nobel Prize in Physics — Nov 1921

The Nobel Prizes are five separate prizes that, according to Alfred Nobel's will of 1895, are awarded to 'those who, during the preceding year, have conferred the greatest benefit to humankind.'

Royal Swedish Academy of Sciences

www.nobelprize.org/prizes/physics/1921/einstein/biographical

Max Planck Medal — 2029

Awarded for outstanding scientific achievement

German Physical Society

Publications

Zur Elektrodynamik bewegter Körper
It concerned an interpretation of the Michelson–Morley experiment and the properties of light and time. Special relativity incorporates the principle that the speed of light is the same for all inertial observers regardless of the state of motion of the source.
Albert Einstein
en.wikisource.org/wiki/Translation:On_the_Electrodynamics_of_Moving_Bodies

Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt
In the second paper, he applied the quantum theory to light to explain the photoelectric effect. In particular, he used the idea of light quanta (photons) to explain experimental results, but stressed the importance of the experimental results. The importance of his work on the photoelectric effect earned him the Nobel Prize in Physics in 1921.
Albert Einstein
de.wikisource.org/wiki/%C3%9Cber_einen_die_Erzeugung_und_Verwandlung_des_Lichtes_betreffenden_heuristischen_Gesichtspur

Die Grundlage der allgemeinen Relativitätstheorie
The publication of the theory of general relativity made him internationally famous. He was professor of physics at the universities of Zurich (1909–1911) and Prague (1911–1912), before he returned to ETH Zurich (1912–1914).
Albert Einstein
de.wikisource.org/wiki/Die_Grundlage_der_allgemeinen_Relativit%C3%A4tstheorie

Skills

Physics

Languages

German
Native speaker

English
Fluent

Interests

Physics

Certificates

Machine Learning Jan 2018

Quantum Computing Jan 2018

Quantum Information Jan 2018

Projects

Quantum Computing Jan 2018 – Jan 2018
Quantum computing is the use of quantum-mechanical phenomena such as superposition and entanglement to perform computation. Computers that perform quantum computations are known as quantum computers.

- Quantum Teleportation
- Quantum Cryptography

References

Professor John Doe

Professor Jane Smith